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# **Homework 3 — Attitude Control of a Launch Vehicle in Atmospheric Flight**

DYNAMICS AND CONTROL OF LAUNCH VEHICLES

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# 1 Introduction

This report addresses the design of an attitude control system for a launch vehicle (LV) during the atmospheric phase of flight, at the instant of maximum dynamic pressure ( $t = 72$  s,  $\max\bar{q}$ ). This is the hardest point to control: the aerodynamic moment destabilises the airframe and the wind disturbance is largest. I design the controller with classical frequency-domain tools, using the Nichols chart as the main stability indicator, following the short-period pitch-plane model of Greensite [1].

## 1.1 Design methodology: frozen coefficients at $\max\bar{q}$

Launch-vehicle attitude control in atmospheric flight is classically handled by the **time-slice** (frozen-coefficient) method: the slowly varying mass, inertia and aerodynamic coefficients are frozen at a representative instant, a linear time-invariant short-period model is built and analysed at that operating point, and the controller is then validated against time-varying simulations [2, Sec. 2]. The main sizing disturbance is the wind profile, which drives both the trajectory dispersion and the structural bending loads—hence the gust analysis of Tasks 1 and 2. I pick  $t = 72$  s because it is the load- and stability-critical slice: the aerodynamic destabilising moment and the wind-induced angle of attack are both near their maxima. The catch is that point-wise Nichols margins certify only the frozen model and do not by themselves guarantee stability of the underlying time-varying system; this is what motivates the parametric robustness study of Task 3 and, eventually, gain scheduling along the trajectory.

## 1.2 Plant model

The pitch-plane short-period dynamics are described by the linear time-invariant model (assignment Eq. 1), with state vector  $\mathbf{x} = [z, \dot{z}, \theta, \dot{\theta}, \eta, \dot{\eta}]^\top$  (lateral drift, pitch attitude and first bending mode), control input  $\delta$  (TVC deflection) and disturbance  $\alpha_w$  (wind angle of attack):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & a_1 & a_1 V + a_4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & A_6/V & A_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega_1^2 & -2\zeta_1\omega_1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ a_3 \\ 0 \\ K_1 \\ 0 \\ -\phi_{1,TVC}T_c \end{bmatrix} \delta + \begin{bmatrix} 0 \\ -a_1 V \\ 0 \\ -A_6 \\ 0 \\ 0 \end{bmatrix} \alpha_w. \quad (1)$$

The pitch channel is the core of the problem and follows from a moment balance about the centre of mass. The aerodynamic moment  $N_\alpha l_\alpha$  acts on the local incidence—built from the pitch attitude  $\theta$ , the plunging term  $\dot{z}/V$  and the wind angle  $\alpha_w$ —while the

gimballed thrust contributes a control moment  $T_c l_c \delta$ ; dividing the balance  $I_{yy} \ddot{\theta} = N_\alpha l_\alpha (\theta + \dot{z}/V - \alpha_w) + T_c l_c \delta$  by  $I_{yy}$  reproduces row 4 of Eq. (1),

$$\ddot{\theta} = A_6 \theta + \frac{A_6}{V} \dot{z} + K_1 \delta - A_6 \alpha_w, \quad (2)$$

and identifies the two structural coefficients: the **aerodynamic moment**  $A_6 = N_\alpha l_\alpha / I_{yy} > 0$  (denoted  $\mu_\alpha$ ) and the **control effectiveness**  $K_1 = T_c l_c / I_{yy}$  ( $\mu_c$ ). With  $A_6 > 0$  the airframe transfer function  $\theta/\delta = K_1/(s^2 - A_6)$  carries an open-loop pole at  $s = +\sqrt{A_6} = +1.84$  rad/s in the right half-plane: the **statically unstable** airframe of a vehicle whose centre of pressure lies ahead of its centre of mass, for which feedback is mandatory [3, Eqs. (9.2.25)–(9.2.26)]. The TVC coefficient  $K_1$  sets the control authority available to counter it.

The sensed quantities are produced by the inertial unit (assignment Eq. 2), which couples the bending coordinate  $\eta$  into the gyro and accelerometer channels through  $\sigma_{1,ins}$  and  $\phi_{1,ins}$ :

$$\theta_m = \theta + \sigma_{1,ins} \eta, \quad \dot{\theta}_m = \dot{\theta} + \sigma_{1,ins} \dot{\eta}, \quad z_m = z - \phi_{1,ins} \eta, \quad \dot{z}_m = \dot{z} - \phi_{1,ins} \dot{\eta}. \quad (3)$$

This contamination is exactly what makes the lightly damped first bending mode—written with  $\omega_1 \equiv \omega_{BM} = 18.9$  rad/s and  $\zeta_1 \equiv \zeta_{BM} = 0.005$  in Eq. (1)—visible to the controller and drives the need for a dedicated filter (Task 2).

The actuator is the second-order TVC of Eq. 3,

$$W_{TVC}(s) = \frac{\omega_{TVC}^2}{s^2 + 2\zeta_{TVC}\omega_{TVC}s + \omega_{TVC}^2}, \quad \omega_{TVC} = 70 \text{ rad/s}, \quad \zeta_{TVC} = 0.7, \quad (4)$$

in series with a pure transport delay  $\tau = 20$  ms. The delay is approximated by a third-order Padé rational function  $e^{-\tau s} \approx P_3(s)$ , the diagonal [3/3] member of the family obtained by matching the Taylor expansion of  $D(s)e^{-\tau s} = N(s)$  to the highest possible order; the illustrative second-order form is

$$e^{-\tau s} \approx \frac{12 - 6\tau s + (\tau s)^2}{12 + 6\tau s + (\tau s)^2}, \quad (5)$$

an **all-pass** rational function ( $|P(j\omega)| = 1$  for all  $\omega$ ) that reproduces the phase lag of the delay while leaving the loop gain unchanged. I keep order three, matching the implementation, so that the approximation stays accurate up to the bending frequency, where  $\omega_{BM}\tau \approx 0.38$  is no longer negligible [4]. Parameters at  $t = 72$  s are read from the reference data set **GreensiteLPV\_DATA.mat** (cross-checked against Table 1 of the assignment); the salient values are reported in Table 1.

Table 1: Pitch-plane LV parameters at the max- $\bar{q}$  condition ( $t = 72$  s).

| Quantity              | Symbol                                            | Value                                   |
|-----------------------|---------------------------------------------------|-----------------------------------------|
| Aerodynamic moment    | $A_6 = \mu_\alpha$                                | $3.382 \text{ s}^{-2}$                  |
| Control effectiveness | $K_1 = \mu_c$                                     | $4.565 \text{ s}^{-2}$                  |
| Drift coefficients    | $a_1, a_3, a_4$                                   | $-0.0154, 20.61, -27.27 \text{ s}^{-2}$ |
| Relative velocity     | $V$                                               | $937.7 \text{ m/s}$                     |
| Bending mode          | $\omega_{BM}, \zeta_{BM}$                         | $18.9 \text{ rad/s}, 0.005$             |
| INS coupling          | $\phi_{1,ins} [-], \sigma_{1,ins} [\text{rad/m}]$ | $0.8, 0.178$                            |
| TVC actuator          | $\omega_{TVC}, \zeta_{TVC}, \tau$                 | $70 \text{ rad/s}, 0.7, 20 \text{ ms}$  |

### 1.3 Control architecture

A proportional–derivative attitude law is adopted, augmented by a weak feedback on the lateral-drift channel:

$$\delta_{cmd} = K_{P,\theta} (\theta_{ref} - \theta_m) - K_{D,\theta} \dot{\theta}_m - K_{P,z} z_m - K_{D,z} \dot{z}_m, \quad (6)$$

with  $K_{P,z}, K_{D,z}$  small and negative, of order  $10^{-3}$ , as recommended in the assignment guidelines. Closing the loop also on  $(z_m, \dot{z}_m)$  trades between two classical objectives in wind: containing the lateral drift (drift minimum) and containing the angle of attack  $\alpha = \theta + \dot{z}/V + \alpha_w$ , hence the structural load  $\bar{q}\alpha$  (load minimum) [5, Lecture 17] — the binding concern at the max- $\bar{q}$  condition studied here. The weak negative gains keep the drift bounded without fighting the natural load-relieving weathercock of the airframe; I therefore monitor both  $z$  and  $\bar{q}\alpha$  in the gust simulations. The actual deflection follows from the actuator and the bending filter  $H_x(s)$  (Task 2):  $\delta = W_{TVC}(s) e^{-\tau s} H_x(s) \delta_{cmd}$ .

### 1.4 A note on conditional stability

Because the open-loop airframe carries a single right-half-plane pole ( $P_R = 1$ , the  $+1.84 \text{ rad/s}$  aerodynamic pole), the loop is **conditionally stable**. By the Nyquist criterion  $N = P_R - Z_R$ , closed-loop stability ( $Z_R = 0$ ) demands exactly one counter-clockwise encirclement of the critical point, so stability is lost if the loop gain is reduced below a threshold, not only if it is increased [3, Eq. (9.4.12), Figs. 9.17–9.18]. An aerodynamically unstable vehicle driven through a second-order (or higher) actuator therefore has two gain margins—a low-frequency aerodynamic margin (reduce gain  $\rightarrow$  unstable) and a high-frequency rigid-body margin (increase gain  $\rightarrow$  unstable)—which on the Nichols chart show up as two 0 dB crossings of the  $-180^\circ$  line. I therefore read the `margin` routine in magnitude, and the assignment targets  $|GM| \approx 6 \text{ dB}$  and  $|PM| \approx 30^\circ$  refer to this finite stable gain band. This is standard practice for the attitude control of aerodynamically unstable launchers [6, 4].

## 2 Task 1 — Rigid Launch Vehicle

Figure 1 sketches the design pipeline of `main_task1.m`, which Tasks 2 and 3 reuse with richer plant and actuator models.

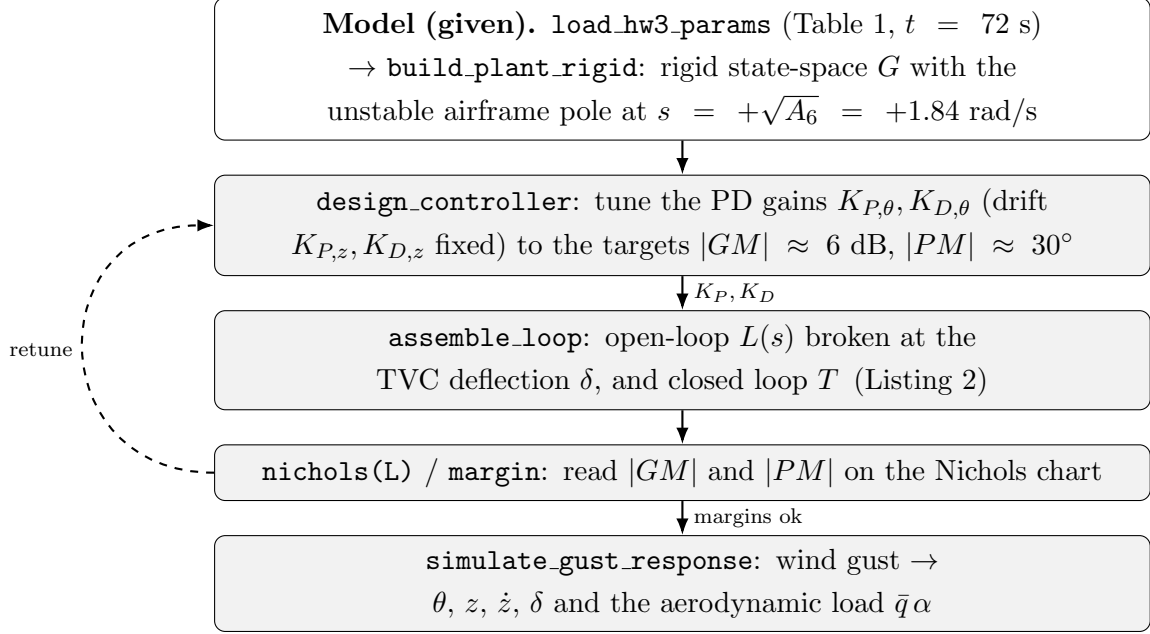


Figure 1: Task 1 design pipeline (`main_task1.m`). Tasks 2 and 3 reuse it with the full flexible plant and the TVC+delay+notch chain `Wact`. Listing numbers refer to Appendix A.

### 2.1 Rigid model and loop

Neglecting the bending mode, the plant reduces to the four states  $[z, \dot{z}, \theta, \dot{\theta}]$  (`build_plant_rigid.m`, Listing 1) and the actuator is taken as ideal:  $W_{TVC}(s) = 1$  and no transport delay. The open-loop transfer broken at the TVC deflection, in the  $1 + L$  convention, is obtained by closing the static PD law of Eq. (6) around the plant (`assemble_loop`, Listing 2):

$$L(s) = K_{P,\theta} G_{\delta \rightarrow \theta}(s) + K_{D,\theta} G_{\delta \rightarrow \dot{\theta}}(s) + K_{P,z} G_{\delta \rightarrow z}(s) + K_{D,z} G_{\delta \rightarrow \dot{z}}(s), \quad (7)$$

where  $G_{\delta \rightarrow \bullet}$  are the plant channels from  $\delta$  to each measurement. The dominant term is the pitch loop  $L_\theta(s) = (K_{P,\theta} + K_{D,\theta}s) K_1 / (s^2 - A_6)$ , whose closed-loop characteristic polynomial  $s^2 + K_1 K_{D,\theta} s + (K_1 K_{P,\theta} - A_6)$  is stable only for  $K_{P,\theta} > A_6 / K_1 = 0.74$  and  $K_{D,\theta} > 0$ , so the proportional gain has to overcome the aerodynamic instability.

### 2.2 Tuning

The pitch gains were tuned (by a small `fminsearch` wrapper around the margin computation) to meet the assignment targets, while the lateral gains were fixed at

$K_{P,z} = K_{D,z} = -10^{-3}$ . The resulting design is

$$K_{P,\theta} = 1.98, \quad K_{D,\theta} = 1.40, \quad K_{P,z} = K_{D,z} = -10^{-3}, \quad (8)$$

with margins  $|GM| = 6.0$  dB at 0.34 rad/s and  $|PM| = 30^\circ$  at 0.21 rad/s. In the pole-placement language of the course notes [5, Lecture 15] the retained gains place the dominant pitch pair at  $\omega_c = \sqrt{K_1 K_{P,\theta} - A_6} = 2.4$  rad/s — inside the usual 1–4 rad/s control band — with an equivalent damping  $\zeta = K_1 K_{D,\theta} / (2\omega_c) = 1.34$ . That sits above the 0.7–0.8 pole-placement range because here the damping is set by the phase-margin target, not by a step-response specification.

These relations invert the closed-loop characteristic polynomial: matching  $s^2 + K_1 K_{D,\theta} s + (K_1 K_{P,\theta} - A_6)$  to  $s^2 + 2\zeta\omega_c s + \omega_c^2$  gives the proportional–derivative design equations

$$K_{P,\theta} = \frac{\omega_c^2 + A_6}{K_1}, \quad K_{D,\theta} = \frac{2\zeta\omega_c}{K_1}, \quad (9)$$

the form tabulated by Barrows and Orr for TVC attitude control [3] (Eqs. (9.2.37) and (9.2.38)). The same reference gives a closed-form estimate of the low-frequency aerodynamic gain margin,  $GM_a \approx 20 \log_{10}(K_1 K_{P,\theta} / A_6) = 20 \log_{10}(4.565 \times 1.98 / 3.382) \approx 8.5$  dB [3, Eq. (9.2.39)]; this single-channel figure sits a couple of dB above the  $|GM| = 6.0$  dB returned by `margin` on the full four-state loop, the difference coming from the weak drift channel and the conditionally-stable low-frequency reading. The gains themselves came from the `fminsearch` wrapper minimising the equal-weight margin miss  $(|GM| - 6)^2 + (|PM| - 30)^2$  from the initial guess [2.0, 1.4], so the reported (1.98, 1.40) is the local minimiser of that objective. The Nichols chart of Figure 2 shows the curve passing between the two critical points  $(-180^\circ, 0$  dB) and  $(-540^\circ, 0$  dB) — which is what conditional stability looks like, the two crossings being the  $360^\circ$  wraps of the single-encirclement Nyquist condition  $N = P_R = 1$  of Section 1 — and clearing the 6 dB closed-loop contour; the markers locate the phase and gain crossovers where the margins are read.

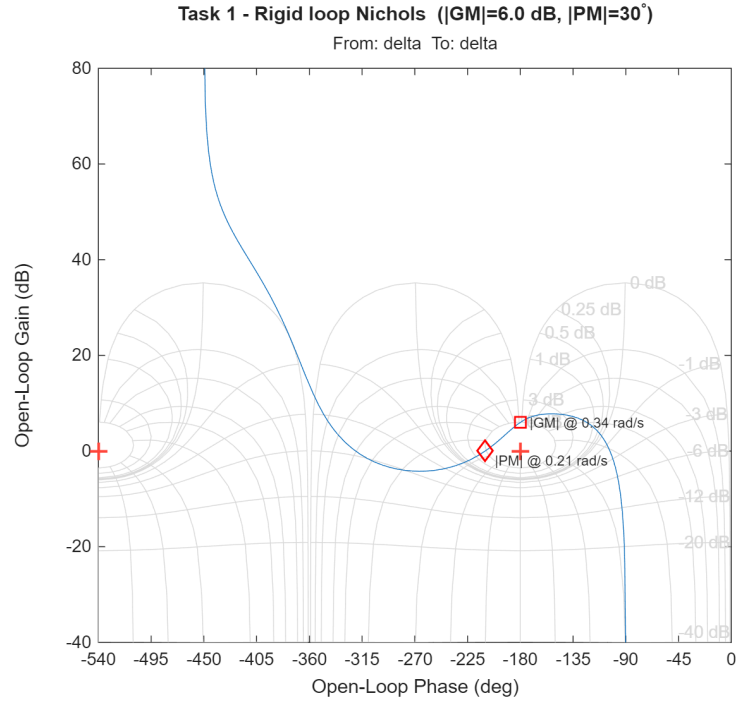


Figure 2: Task 1 — Nichols chart of the rigid open-loop transfer. The loop is conditionally stable; the design meets  $|GM| \approx 6$  dB and  $|PM| \approx 30^\circ$  (markers at the crossover frequencies).

## 2.3 Wind-gust response

The closed loop is excited by a deterministic “1-cosine” wind gust whose amplitude ( $V_g = 6.4$  m/s) is taken from the reference dry-wind dispersion (`drywind.mat`, severe regime) at the current altitude, converted to a wind angle of attack  $\alpha_w = v_w/V$  with a peak of  $0.39^\circ$ . The four key time histories requested by the assignment are shown in Figure 3: the pitch attitude peaks at  $0.139^\circ$ , the TVC deflection at  $0.38^\circ$ , and the lateral drift builds up smoothly to 2.5 m, which is what the weak drift feedback gives. The response is well damped and the actuator effort stays small.



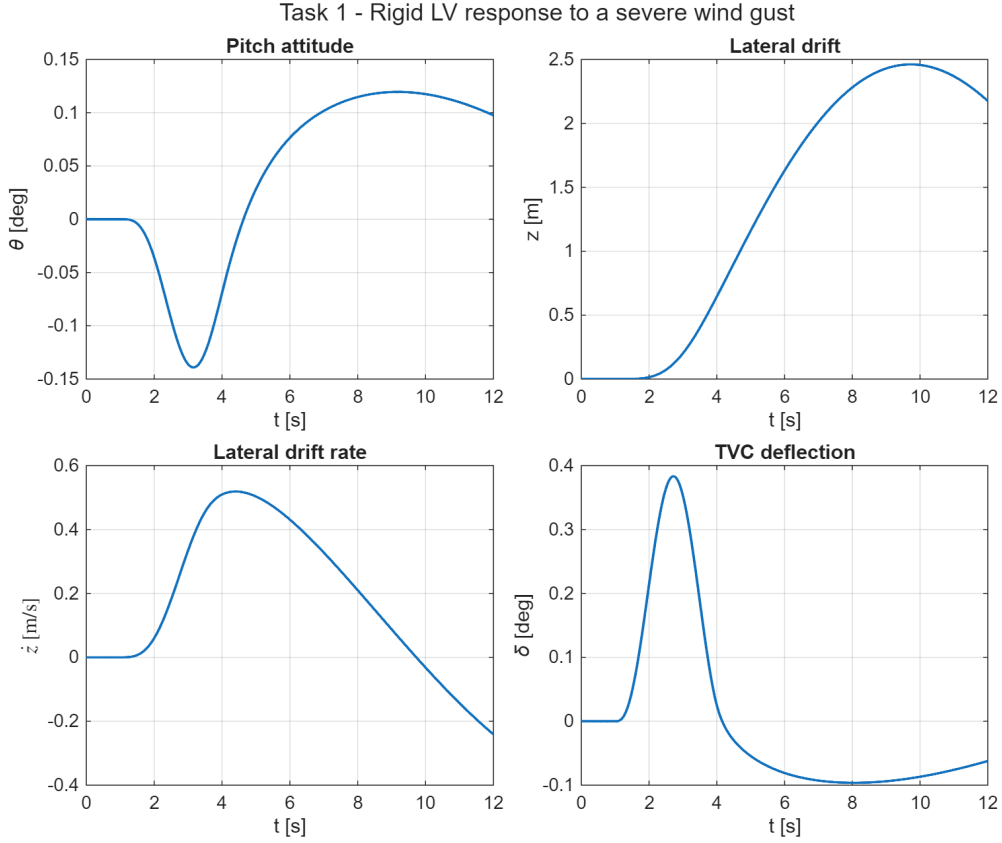


Figure 3: Task 1 — response to a severe wind gust: pitch attitude  $\theta$ , lateral drift  $z$ , drift rate  $\dot{z}$  and TVC deflection  $\delta$ .

Since  $\max \bar{q}$  is the load-critical point of the trajectory, the total angle of attack  $\alpha = \theta + \dot{z}/V + \alpha_w$  and the structural-load indicator  $\bar{q}\alpha$  are also monitored (Figure 4). The attitude loop pitches the vehicle slightly into the wind ( $\theta \rightarrow -0.14^\circ$  at the gust peak), so the peak total incidence,  $0.31^\circ$ , ends up below the wind contribution alone — a mild load-relieving action. With  $\bar{q} = 81$  kPa the peak load indicator is  $\bar{q}\alpha = 25$  kPa deg, the incidence remaining an order of magnitude below the few-degree load limit typical of a slender launcher, so at this gust level the drift-minimum-oriented gain choice does not conflict with the load-minimum requirement.

This drift-versus-load tension is the standard trade in launch-vehicle wind control [2, Sec. 3.1]. The characteristic equation of the rigid airframe in wind is a cubic with two oscillatory rigid-body roots plus one real drift (flight-path) root; the final-value theorem applied to a step gust gives a steady drift rate  $\dot{z}_{ss} = -V\alpha_w$ , independent of the gains — which is why the lateral drift of Figure 3 accumulates smoothly rather than being nulled. Two limiting cases bracket the design. The **drift-minimum** condition (drift-channel coefficient  $B_0 = 0$ ) holds the trajectory dispersion down. The **load-minimum** condition ( $K_{P,\theta} = 0$ , so the steady incidence  $\alpha_{ss} = 0$ ) minimises  $\bar{q}\alpha$ , but pays for it with a slow unstable drift mode. The weak negative drift gains I used here are a small step toward the drift-minimum end of this trade.

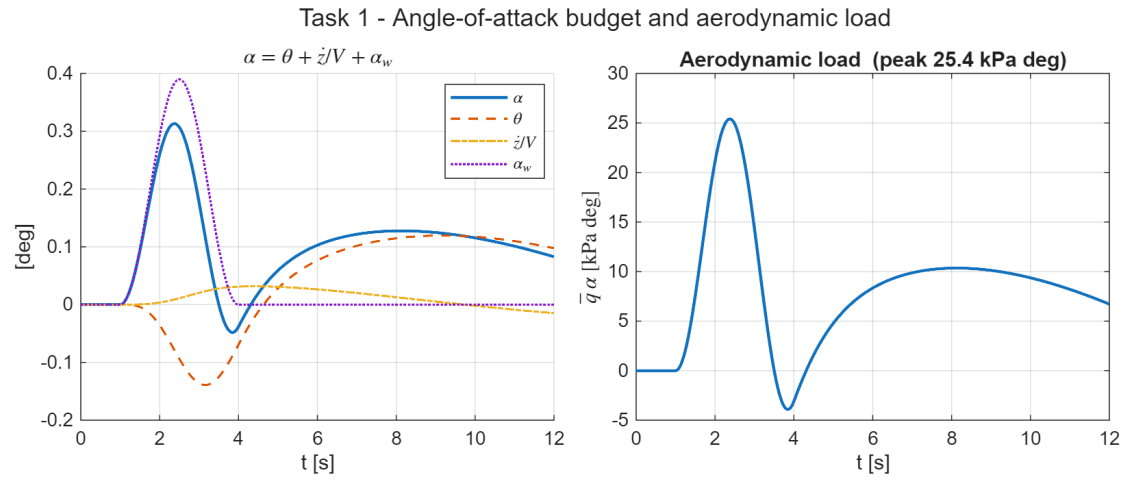


Figure 4: Task 1 — angle-of-attack budget  $\alpha = \theta + \dot{z}/V + \alpha_w$  (left) and aerodynamic load indicator  $\bar{q} \alpha$  (right) during the severe gust.

## 3 Task 2 — Full Launch Vehicle Model

Task 2 reintroduces the full dynamics of Eq. (1): the lightly damped first bending mode, the second-order TVC actuator of Eq. (4) with its 20 ms delay, and the INS measurement model of Eq. (3). I built the design up incrementally, as the assignment guidelines suggest, keeping the Task 1 PD gains fixed throughout. The guideline note says retuning is usually unnecessary, and that holds here.

### 3.1 Step B — uncompensated full model

Adding the TVC, delay and bending mode to the Task 1 controller leaves the low-frequency margins almost untouched, but the bending resonance shows up in the loop at  $|L(\omega_{BM})| = +39$  dB. With a damping of only  $\zeta_{BM} = 0.005$  this peak encircles the critical point and the closed loop goes **unstable** (a closed-loop pole at  $\text{Re}(s) = +5.6$ ). The red curve in Figure 5 shows the bending loop crossing into the unstable region.

### 3.2 Step C — bending filter trade study

The guideline filter of Eq. 4 admits two readings: as printed (negative numerator damping, a non-minimum-phase **lead-lag** that phase-stabilises the resonance by rotating it on the Nichols chart) or in its minimum-phase variant (a band-reject **notch** that gain-stabilises the resonance by driving it below 0 dB) [5, Lecture 18]. Instead of picking one up front, I compare four candidates with the same fixed PD gains (all built by `build_notch_filter.m`, Listing 3); Table 2 collects the resulting margins (rigid-body and minimum gain margin, phase margin, delay margin on top of the 20 ms already modelled, and the residual loop gain at the bending frequency).

- **C-LL — Eq. 4 lead-lag alone.** The filter is swept over the ranges suggested in the assignment ( $\zeta_{x,N} \in [0.1, 0.3]$ ,  $\zeta_{x,D} \in [0.4, 0.6]$ ,  $\omega_x = \omega_{BM} \pm 4$  rad/s, 75 combinations): none stabilises the loop. The least-unstable combination ( $\omega_x = 18.9$ ,  $\zeta_{x,N} = 0.10$ ,  $\zeta_{x,D} = 0.60$ ) still leaves a closed-loop pole at  $\text{Re}(s) = +0.16$  — the green curve in Figure 5 shows its bending lobe still pinned to the critical region. Phase rotation alone cannot rescue a +39 dB resonance here.
- **C-N — deep minimum-phase notch.** Gain stabilisation needs more than 39 dB of attenuation at  $\omega_{BM}$ , which forces the numerator damping far below the suggested range:  $\zeta_{x,N} = 0.002$ ,  $\zeta_{x,D} = 0.7$  give a  $-51$  dB null and drive  $|L(\omega_{BM})|$  to  $-12$  dB. The depth follows the closed-form attenuation of a second-order notch,  $K_{\text{notch}} = 20 \log_{10}(\zeta_{x,N}/\zeta_{x,D})$ , so that  $20 \log_{10}(0.002/0.7) \approx -51$  dB at  $\omega_{BM}$  [7, Eq. (3.18)]; pulling a +39 dB peak below 0 dB requires  $\zeta_{x,N}/\zeta_{x,D} \leq 10^{-39/20} \approx 0.011$ , i.e.  $\zeta_{x,N} \lesssim 0.008$  with  $\zeta_{x,D} = 0.7$ , which is why the sizing

has to leave the assignment’s suggested  $\zeta_{x,N} \in [0.1, 0.3]$  range. I keep the null narrow because the bending frequency sits only a decade above the rigid crossover ( $\omega_{BM}/\omega_c \approx 8$ ): a wide band-reject section would leak phase lag into the conditionally stable low-frequency region.

- **C-T — notch triplet.** The notch-triplet recipe against bending-frequency uncertainty places three cascaded notches at  $\{0.9, 1.0, 1.1\} \omega_{BM}$  [7, Eq. (3.19), Fig. 3.11]. Sized to the same depth, however, the three wide ( $\zeta_{x,D} = 0.7$ ) sections pile up about  $30^\circ$  of phase lag at the rigid crossover and the conditionally stable loop goes unstable at nominal ( $|PM|$  collapses to  $11.6^\circ$ ). This matches the course view: the triplet belongs to the higher, well-separated modes, while the first mode — too close to the control band — has to be handled either with an exact notch or by phase stabilisation.
- **C-NLL — notch + lead-lag.** Reading the guideline literally (“the Notch filter and possibly other filters”), the deep notch is cascaded with the Eq. 4 lead-lag, the partner swept over the same ranges: 31/75 combinations now stabilise the loop. The best one ( $\omega_x = 22.9$ ,  $\zeta_{x,N} = 0.10$ ,  $\zeta_{x,D} = 0.40$ , selected for maximum delay margin) is stable but with clearly thinner nominal margins ( $|PM| = 9.2^\circ$ ,  $\min |GM| = 0.7$  dB). Its payoff shows up in the robustness analysis below.

Table 2: Task 2 — bending-filter trade with fixed Task-1 PD gains.

| Candidate        | rigid $ GM $ [dB] | min $ GM $ [dB] | $ PM $ [ $^\circ$ ] | DM [ms] | $ L(\omega_{BM}) $ [dB] | stable     |
|------------------|-------------------|-----------------|---------------------|---------|-------------------------|------------|
| B (no filter)    | 6.34              | 6.34            | 30.8                | —       | +38.8                   | no         |
| C-LL (lead-lag)  | 6.58              | 0.61            | 9.0                 | —       | +23.2                   | no         |
| C-N (deep notch) | 6.58              | 1.81            | 26.5                | 58.8    | −12.1                   | <b>yes</b> |
| C-T (triplet)    | 7.07              | 1.48            | 11.6                | —       | −46.0                   | no         |
| C-NLL (notch+LL) | 6.72              | 0.73            | 9.2                 | 21.9    | −18.3                   | yes        |

I keep the **deep notch**: it gain-stabilises the resonance while preserving the rigid-body margins ( $|GM| = 6.6$  dB,  $|PM| = 26.5^\circ$  — a few degrees below the  $30^\circ$  of the ideal-actuator loop of Task 1, the difference being the phase lag of the second-order TVC and the 20 ms transport delay), leaves a 58.8 ms delay margin, and needs no retuning of the PD gains, in line with the guideline note. The bending mode itself stays at its lightly damped but stable open-loop location. One caveat: the minimum gain margin of the stabilised loop, set by the bending shoulder, is 1.8 dB — fine for this exercise but well below the  $\geq 12$  dB of gain margin (and  $45^\circ$  of phase margin) usually required of gain-stabilised bending modes, as opposed to the 6 dB/ $30^\circ$  rigid-body figure [3, p. 279]. The tougher bending requirement comes from the larger modelling uncertainty and latency sensitivity at high frequency. Getting the deeper margin would cost a stiffer notch or a retuned loop gain.

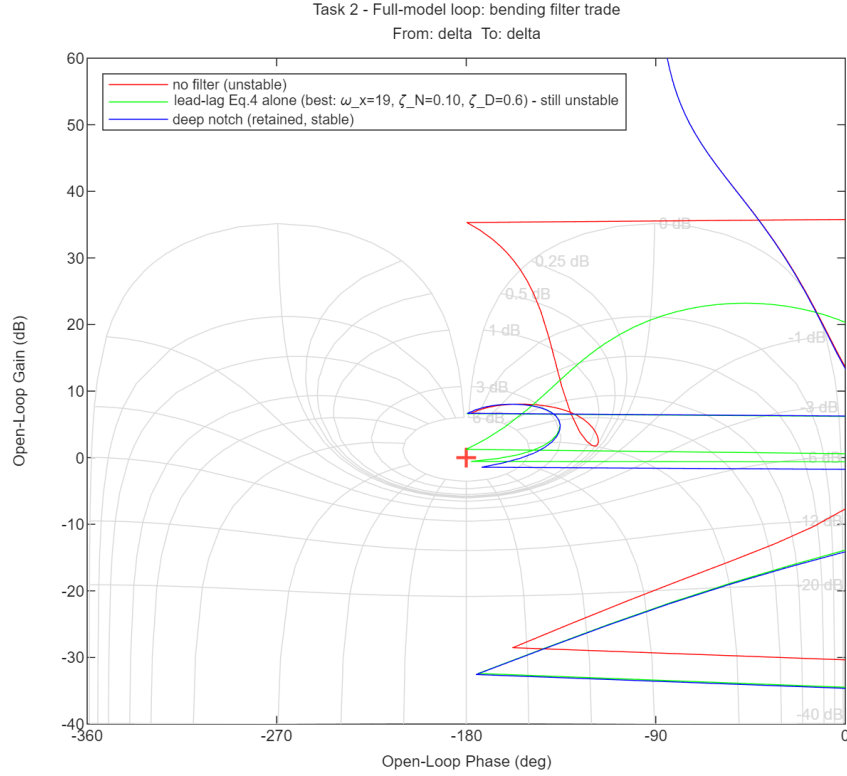


Figure 5: Task 2 — bending-filter trade on the Nichols chart (phase wrapped): without compensation the +39 dB resonance destabilises the loop (red); the best Eq.-4 lead-lag alone rotates the bending lobe but cannot clear the critical region (green); the deep notch gain-stabilises it and restores the rigid margins (blue).

### 3.3 Sensitivity to the bending-frequency knowledge

The deep notch pays for its performance with a  $-51$  dB null only  $0.08$  rad/s wide, i.e. it assumes  $\omega_{BM}$  is known exactly — true in this exercise, where the LPV dataset provides it, but never true in practice, which is why the course notes prescribe the  $\pm 10\%$  triplet. Table 3 makes the trade explicit: each filter is held fixed while the true plant bending frequency is perturbed. The notch tolerates barely  $-5\%$ ; the notch+lead-lag combination, by phase-stabilising whatever leaks past the detuned notch, recovers nearly the full  $\pm 10\%$  band at the cost of the thin nominal margins seen above. In a flight design, where  $\omega_{BM}$  drifts with propellant depletion, the robust variant (or a retuned phase-stabilised loop) would be mandatory; for this max- $\bar{q}$  snapshot I keep the exact notch.

Table 3: Task 2 — closed-loop stability with the filters fixed and the true bending frequency perturbed by up to  $\pm 10\%$ .

| Candidate              | $0.90\omega_{BM}$ | $0.95\omega_{BM}$ | $\omega_{BM}$ | $1.05\omega_{BM}$ | $1.10\omega_{BM}$ |
|------------------------|-------------------|-------------------|---------------|-------------------|-------------------|
| C-N — deep notch       | no                | yes               | yes           | no                | no                |
| C-T — notch triplet    | no                | no                | no            | no                | no                |
| C-NLL — notch+lead-lag | no                | yes               | yes           | yes               | yes               |

### 3.4 Validation and comparison with the rigid model

The gust response of the stabilised full model is shown in Figure 6; all four assignment time histories remain well damped, with peaks  $\theta = 0.132^\circ$ ,  $z = 2.37$  m,  $\delta = 0.39^\circ$  and a load indicator  $\bar{q}\alpha = 24.9$  kPa deg, just below the rigid case. Because the notch is localised at the bending frequency and the TVC bandwidth (70 rad/s) is well above the control crossover, the full-model response is almost indistinguishable from the rigid one (Figure 7): the added dynamics affect stability robustness (the bending and delay margins) far more than nominal performance.

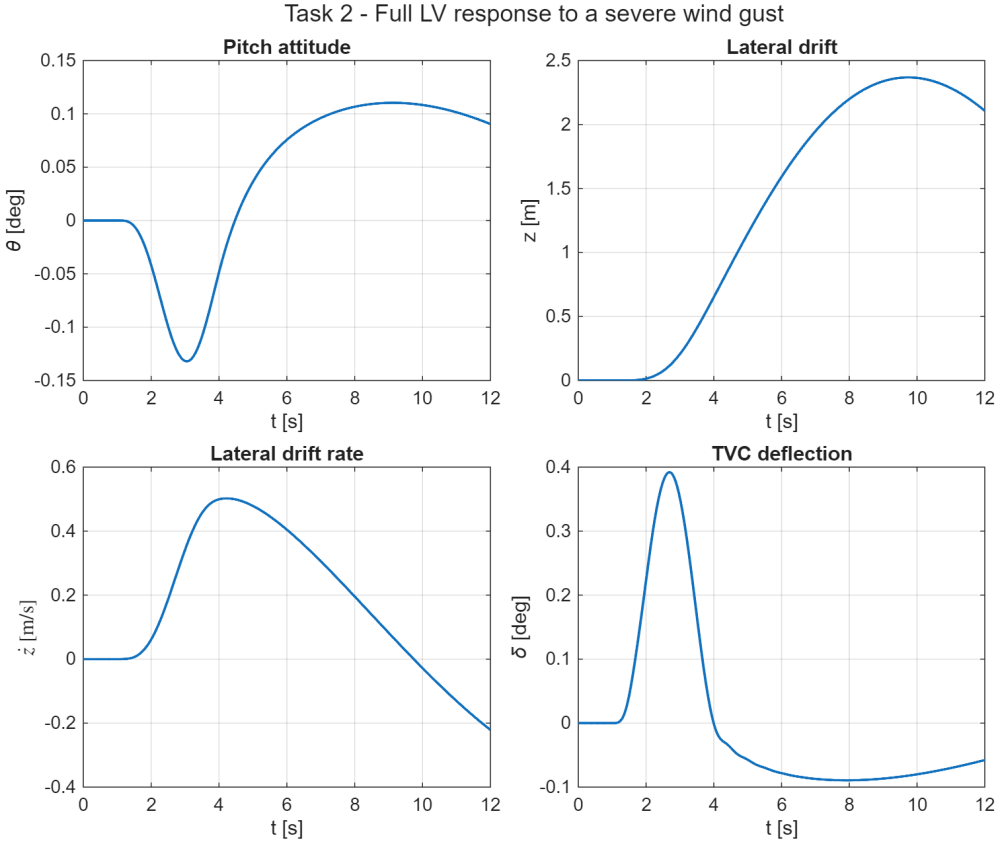


Figure 6: Task 2 — full-model response to the severe wind gust: pitch attitude  $\theta$ , lateral drift  $z$ , drift rate  $\dot{z}$  and TVC deflection  $\delta$ .

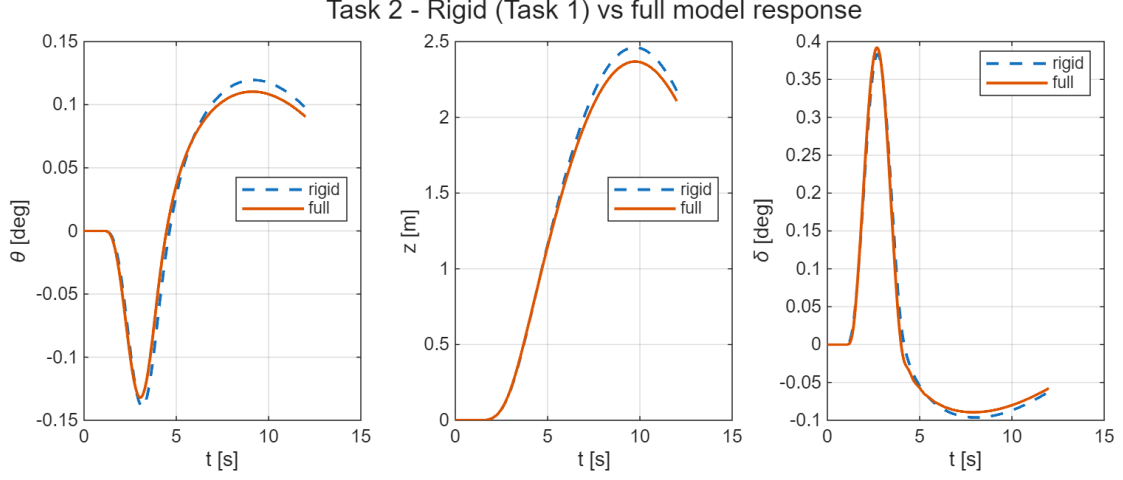


Figure 7: Task 2 — rigid (Task 1) versus full-model gust response: pitch attitude, lateral drift and TVC deflection are nearly coincident.

### 3.5 A note on the tail-wags-dog zero

A second flexible effect bounds what any filter can achieve. The gimbaled nozzle has its own inertia, so its angular acceleration reacts back on the airframe: at the **tail-wags-dog** (TWD) frequency

$$\omega_{TWD}^2 = -\frac{F_R X_G}{I_{EG} + X_G m_E l_E}, \quad (10)$$

nozzle deflection produces zero net vehicle response, a transmission zero that caps the achievable closed-loop bandwidth [3, Eq. (6.1.11)]. The  $-\phi_{1,TVC} T_c$  entry in the bending-forcing column of Eq. (1) is the flexible-mode analogue of this nozzle-reaction term. My model treats the actuator as a stiff second-order block (Eq. (4)) and so neglects the nozzle back-reaction — the dog-wags-tail feedback, which would add the gimbal angle as an extra state. A higher-fidelity design would keep it, one of the extensions noted in the conclusions.

## 4 Task 3 — Robustness to Parametric Uncertainty

The aerodynamic moment  $\mu_\alpha = A_6$  and the control effectiveness  $\mu_c = K_1$  are treated as uncertain. The controller designed in Task 2 (PD gains plus bending notch) is held **fixed** and I assess its robustness over the four corner cases of the uncertainty box, i.e. the vertices V1–V4 obtained by independently varying  $\mu_\alpha$  and  $\mu_c$  by  $\pm 30\%$ . Since each vertex perturbs both parameters at once, I also evaluate the four one-at-a-time variations S1–S4 ( $\pm 30\%$  on a single parameter): they sit on the edge midpoints of the box rather than on its corners, but they attribute the observed effects to each parameter.

### 4.1 Frequency- and time-domain analysis

Table 4 collects, for every case, the rigid-body (low-frequency) gain margin, the smallest gain margin over the whole loop (governed by the bending shoulder), the phase margin, the delay margin, and the peak pitch and lateral-drift excursions to the same severe gust. All vertices stay stable: the fixed Task-2 controller tolerates the entire  $\pm 30\%$  uncertainty box without re-design.

Table 4: Task 3 — margins and gust-response peaks: box vertices (V1–V4, the assignment corner cases) and one-at-a-time sensitivities (S1–S4).

| Case    | $\mu_\alpha$ | $\mu_c$ | rigid $ GM $ [dB] | min $ GM $ [dB] | $ PM $ [°] | DM [ms] | peak $\theta$ [°] | peak $z$ [m] |
|---------|--------------|---------|-------------------|-----------------|------------|---------|-------------------|--------------|
| Nominal | 1.00         | 1.00    | 6.58              | 1.81            | 26.5       | 58.8    | 0.132             | 2.37         |
| V1      | 0.70         | 0.70    | 6.42              | 2.81            | 31.3       | 134.9   | 0.136             | 2.38         |
| V2      | 0.70         | 1.30    | 11.68             | 0.88            | 11.3       | 19.2    | 0.061             | 1.67         |
| V3      | 1.30         | 0.70    | <b>1.53</b>       | 1.53            | <b>9.1</b> | 156.4   | <b>0.353</b>      | <b>4.20</b>  |
| V4      | 1.30         | 1.30    | 6.66              | 0.93            | 12.3       | 21.3    | 0.130             | 2.36         |
| S1      | 0.70         | 1.00    | 9.44              | 1.79            | 25.7       | 56.1    | 0.085             | 1.91         |
| S2      | 1.30         | 1.00    | 4.47              | 1.83            | 22.1       | 61.8    | 0.187             | 2.88         |
| S3      | 1.00         | 0.70    | 3.59              | 2.82            | 18.6       | 144.6   | 0.219             | 3.16         |
| S4      | 1.00         | 1.30    | 8.79              | 0.91            | 11.8       | 20.2    | 0.093             | 2.01         |

The Nichols overlay over the nominal case and the four vertices (Figure 8) shows the curves clustering at low frequency and fanning out near the bending frequency; the time-domain overlay (Figure 9) shows the corresponding spread of the gust response.

### 4.2 Discussion

The sensitivities S1–S4 attribute the effects:  $\mu_\alpha$  moves the unstable airframe pole ( $\pm 30\%$  shifts the rigid margin by roughly  $\mp 2$ – $3$  dB and scales the gust excursions), while  $\mu_c$  scales the whole loop gain (+30% improves the rigid margin but pushes



the gain-stabilised bending shoulder 2.3 dB closer to 0 dB — the leading-order  $20 \log_{10} 1.3 = 2.28$  dB loop-gain shift, only approximate because  $K_1$  also scales the bending-forcing column of Eq. (1) and not merely the loop gain — and halves the delay margin). The vertices then combine these mechanisms:

- **V3** ( $\mu_\alpha \uparrow, \mu_c \downarrow$ ) is the worst case for the rigid loop and for performance, because the two effects compound: the airframe is more unstable and the authority to fight it is reduced at the same time. The rigid gain margin collapses to 1.5 dB with  $|PM| = 9.1^\circ$ , and the gust excursions grow to  $0.35^\circ$  in pitch and 4.2 m of drift — roughly 60% beyond the worst one-at-a-time case (S3). This is why I test the actual vertices and not only the single-parameter variations.
- **V2 and V4** ( $\mu_c \uparrow$ ) are the worst cases for the bending channel: the loop-gain increase squeezes the minimum gain margin to 0.9 dB and the delay margin to  $\sim 20$  ms. The binding constraint in these corners is the flexible side of the loop, not the rigid one.
- **V1** ( $\mu_\alpha \downarrow, \mu_c \downarrow$ ) is benign on both counts: the two reductions nearly cancel in the rigid loop and relax the bending shoulder.

So performance degradation is driven by  $\mu_\alpha$  and reduced by  $\mu_c$ , while the bending channel loses stability margin as  $\mu_c$  grows, and the worst-case combination V3 leaves single-digit margins. Because the margins are non-monotonic in  $\mu_c$  (it helps the rigid loop while hurting the bending shoulder), the global worst case over the  $\pm 30\%$  box need not sit on a vertex: the four corners and the one-at-a-time sensitivities are only a screening, and a denser grid or Monte-Carlo sweep would be needed to certify an interior worst case. The fixed classical design survives the whole box, but a requirement of, say,  $\geq 3$  dB and  $\geq 10^\circ$  in every corner would already call for either gain scheduling on  $\bar{q}$  or a robust-synthesis retune, the standard routes beyond a single fixed classical design [7]. That is the practical takeaway here.

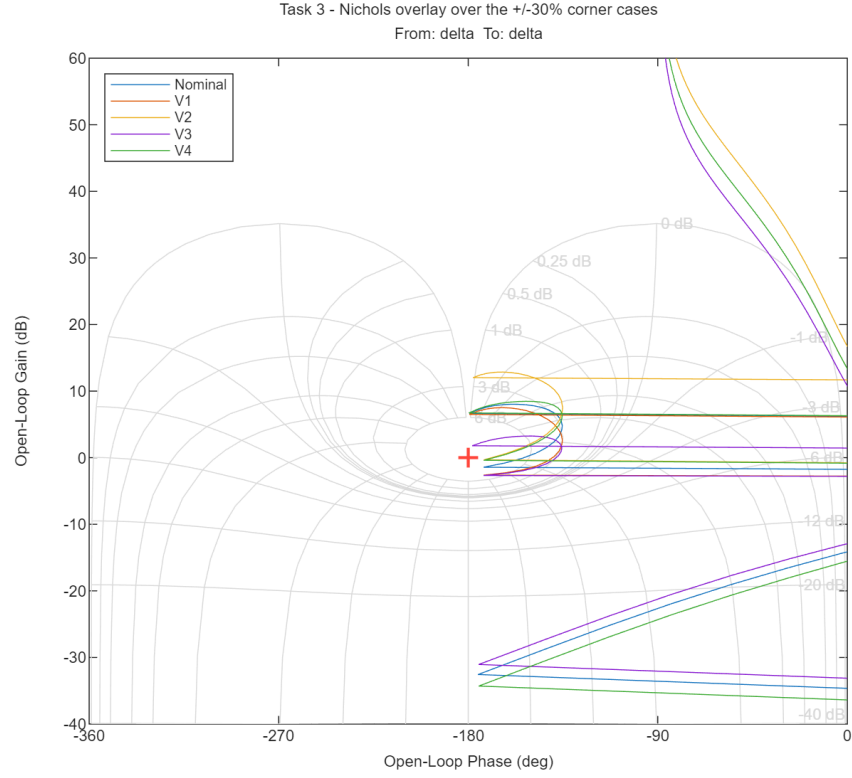


Figure 8: Task 3 — Nichols overlay over the nominal case and the four uncertainty-box vertices.

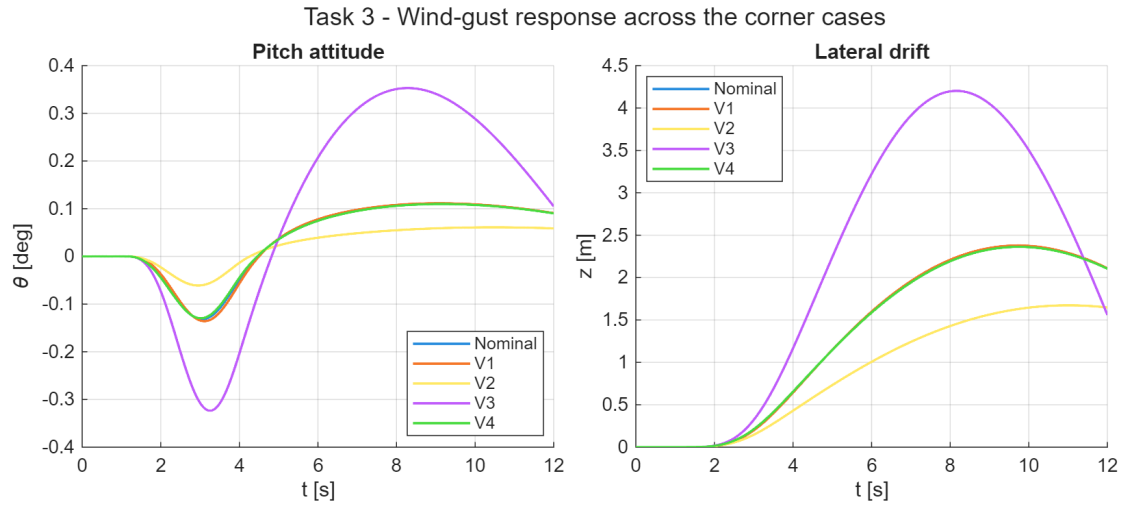


Figure 9: Task 3 — wind-gust response over the nominal case and the four vertices: pitch attitude (left) and lateral drift (right).

## 5 Conclusions

I designed and validated a classical attitude control system for an aerodynamically unstable launch vehicle at the max- $\bar{q}$  condition, working entirely in the frequency domain.

- **Task 1.** A proportional–derivative attitude law with weak negative drift feedback stabilises the rigid airframe (open-loop pole at +1.84 rad/s) and meets the design targets  $|GM| \approx 6$  dB,  $|PM| \approx 30^\circ$ . The loop is conditionally stable, and I kept that in mind when reading the margins.
- **Task 2.** Introducing the TVC actuator, the transport delay and the lightly damped bending mode destabilises the loop through a +39 dB resonance. A four-way filter trade showed that the Eq.-4 lead-lag alone never stabilises the loop within the suggested parameter ranges (0/75 combinations), and that a notch triplet sacrifices too much phase at low frequency for a conditionally stable loop; a deep gain-stabilising notch centred on  $\omega_{BM}$  restores stability while preserving the rigid margins, with negligible change to the nominal gust response. I quantified how much it leans on knowing the bending frequency exactly ( $\pm 5\%$  detuning is fatal) and found a robust notch+lead-lag variant that tolerates nearly  $\pm 10\%$  at the price of thinner nominal margins.
- **Task 3.** The fixed Task 2 controller is robust to  $\pm 30\%$  independent variations of  $\mu_\alpha$  and  $\mu_c$ : all four vertices of the uncertainty box remain stable. Uncertainty on  $\mu_\alpha$  erodes the rigid margin and the gust performance, uncertainty on  $\mu_c$  erodes the bending and delay margins, and the worst vertex (high  $\mu_\alpha$  with low  $\mu_c$ ) compounds them, leaving only 1.5 dB and  $9^\circ$  — the quantitative argument for gain scheduling in a flight design.

**Limitations and extensions.** Four modelling choices bound the scope of this design. (i) The pure proportional–derivative law leaves a steady-state attitude error to a constant gimbal misalignment, which a proportional–integral–derivative extension would null. (ii) The stiff second-order TVC model neglects the nozzle back-reaction (the dog-wags-tail feedback) and the hydraulic load-resonance of the actuator; retaining the gimbal angle as a state, together with the tail-wags-dog zero of Eq. (10) [3, Sec. 6.1], is the natural next fidelity step. (iii) Sensor placement relative to the bending-mode nodes matters as much as the filter — an aft gyro station flips the sign of the modal slope and can phase-stabilise a mode that would otherwise need a notch [2, Sec. 3.3.2.2] — whereas here the INS station is fixed and only the filter is traded. (iv) Propellant sloshing is omitted; it would add low-frequency, lightly damped modes coupling into the rigid loop and the drift channel.

The design shows the textbook trade-off of launch-vehicle attitude control [1, 6, 3]:

enough gain to stabilise the unstable airframe, but not so much that it excites the structural modes, and the Nichols chart lets me see both constraints on one plot. A Simulink block-diagram mirror of the closed loop reproduces the script results and is documented in the accompanying `models/SIMULINK_GUIDE.md`.

## A Implementation: Key Code Excerpts

The full MATLAB sources live in the HM3/ folder of the repository: `main_task1.m`, `main_task2.m` and `main_task3.m` drive the three tasks, and the helper functions (`load_hw3_params`, `build_plant_rigid`, `build_plant_full`, `build_tvc`, `build_notch_filter`, `design_controller`, `assemble_loop`, `simulate_gust_response`) are shared across them. This appendix collects the three routines that set up the control problem: the rigid plant, the loop closure the Nichols analysis is drawn from, and the bending notch.

Listing 1 builds the rigid pitch-plane state-space model at the  $\max\bar{q}$  condition by dropping the bending states from the assignment dynamics; the unstable airframe pole sits at  $+\sqrt{A_6}$ .

```

1 function G = build_plant_rigid(p)
2 % Rigid pitch-plane LV plant at max-qbar (Task 1, Eq. 1 minus bending).
3 % INPUT
4 %   p - param struct (load_hw3_params)
5 % OUTPUT
6 %   G - ss, 4 states [z zdot theta thetadot], in [delta alpha_w],
7 %       out [theta_m thetadot_m z_m zdot_m theta z zdot]
8 %   First 4 outputs feed the controller (= true states here, no bending);
9 %   last 3 are plotting signals.
10
11 A = [0 1 0 0;
12      0 p.a1 p.a1*p.V+p.a4 0;
13      0 0 0 1;
14      0 p.A6/p.V p.A6 0];
15
16 Bd = [0; p.a3; 0; p.K1]; % delta column
17 Bw = [0; -p.a1*p.V; 0; -p.A6]; % alpha_w column
18
19 % Outputs: [theta_m, thetadot_m, z_m, zdot_m, theta, z, zdot]
20 %          state index:  z=1  zdot=2  theta=3  thetadot=4
21 C = [0 0 1 0; % theta_m = theta
22      0 0 0 1; % thetadot_m = thetadot
23      1 0 0 0; % z_m = z
24      0 1 0 0; % zdot_m = zdot
25      0 0 1 0; % theta
26      1 0 0 0; % z
27      0 1 0 0]; % zdot
28 D = zeros(7, 2);
29
30 G = ss(A, [Bd Bw], C, D);
31 G.StateName = {'z', 'zdot', 'theta', 'thetadot'};
32 G.InputName = {'delta', 'alpha_w'};
33 G.OutputName = {'theta_m', 'thetadot_m', 'z_m', 'zdot_m', 'theta', 'z', 'zdot'};
34 G.Name = 'Plant_Rigid';
35 end

```

Listing 1: Rigid pitch-plane LV plant (`build_plant_rigid.m`).

Listing 2 closes the PD attitude and drift loop around the plant, optionally inserting the TVC, delay and notch chain through `Wact`, and returns the SISO open-loop transfer broken at the TVC deflection. This open-loop transfer is what the Nichols charts in

Tasks 1–3 are built from.

```

36
37 % --- open loop at the break point 'delta' (1+L convention) ---
38 L = getLoopTransfer(T, 'delta', -1);
39 L = minreal(tf(L), 1e-6);
40
41 info = struct('Kc', Kc, 'Wact', Wa);
42 end

```

Listing 2: Loop closure and open-loop transfer for the Nichols analysis (assemble\_loop.m).

Listing 3 is the second-order lead-lag / notch section used in Task 2 to stabilize the lightly damped first bending mode.

```

1 function Hx = build_notch_filter(wx, zN, zD, numSign)
2 % Lead-lag / notch section for bending stabilisation (Eq. 4).
3 % Hx(s) = (s^2 + sgn*2*zN*wx*s + wx^2) / (s^2 + 2*zD*wx*s + wx^2)
4 % INPUT
5 % wx - centre frequency [rad/s] (~ wBM +/- 4)
6 % zN - numerator damping (guideline 0.1-0.3)
7 % zD - denominator damping (guideline 0.4-0.6)
8 % numSign - -1 (default, Eq. 4 as printed): RHP zeros, non-min-phase
9 %           phase-shaper; +1: min-phase symmetric notch
10 % OUTPUT
11 % Hx - tf
12
13 arguments
14 wx (1,1) {mustBeNumeric, mustBeReal, mustBePositive}
15 zN (1,1) {mustBeNumeric, mustBeReal, mustBeNonnegative}
16 zD (1,1) {mustBeNumeric, mustBeReal, mustBePositive}
17 numSign (1,1) {mustBeMember(numSign, [-1, 1])} = -1
18 end
19
20 num = [1, numSign*2*zN*wx, wx^2];
21 den = [1, 2*zD*wx, wx^2];
22
23 Hx = tf(num, den);
24 Hx.Name = 'Notch_Hx';
25 end

```

Listing 3: Lead-lag / notch filter for bending stabilization (build\_notch\_filter.m).

## References

- [1] A. L. Greensite. *Analysis and Design of Space Vehicle Flight Control Systems*. Spartan Books, New York, 1970.
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